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Thermodynamic assessment of Cu–Eu and Cu–Yb system

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Abstract

The Cu–Eu and Cu–Yb binary systems have been assessed with CALPHAD method. Liquid, BCC and FCC phases are treated as substitutional solution phases, of which the excess Gibbs energies are modeled by Redlich–Kister polynomial functions. The binary intermetallic compounds are treated as stoichiometric phases. Thermodynamic parameters of various phases have been optimized and the calculated results are in agreement with experimental data.
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Keywords: Thermodynamic calculation; Cu–Eu system; Cu–Yb system; Phase diagram

1. Introduction

Cu-based alloys with high strength and high electronic-conductivity have found wide application in high-speed train and electronic industry. However, high strength is usually incompatible with high electric conductivity in materials. Addition of rare earth elements into copper matrix and adoption of proper heat-treatment may help to achieve high strength while maintaining high conductivity. For example, it was reported that Eu improves the strength and electronic-conductivity while the temperature coefficient of resistivity is negative [1,2]. Eu is also a candidate element for improving the properties, such as magnetic moment, Hall effect and magnetic resistivity, of some Cu-containing magnetic materials [3]. Generally, Eu, Yb are found to be capable of improving resistivity of Cu-based alloys [4,5]. Knowledge of phase diagrams of the related multi-component Cu-alloys is essential to understand the role of these rare earth elements and do help in alloy design (optimal compositions) and fabrication (forming methods and heat-treatment conditions) for improving the material properties of Cu-alloys.

The Cu–Eu and Cu–Yb systems were assessed by Subramanian et al. [6,7]. However, their assessments [6,7] were only based on the phase diagram data and the lattice stabilities for Cu, Eu and Yb were out-of-date [8]. Moreover, there are new thermodynamic data [9] available since the previous calculation. The purpose of this work is then to thermodynamically assess the Cu–Eu and Cu–Yb systems with all the updated data.

2. Evaluation of experimental data

2.1. Cu–Eu system

The melting points of Cu and Eu in this work are accepted as 1084.87 °C [8] and 822 °C [11,12], respectively. Note, the melting point reported by Costa et al. [10] for Eu was 7 °C lower than the accepted value [11,12] due to impurity in their Eu samples (99% Eu). By thermal analysis Costa et al. [10] found that the solid solubility of Cu in Eu is negligible. Similarly, there was no evidence of terminal solid solubility of Eu in Cu. According to Subramanian et al. [6] there are four intermediate phases in this system: The Cu_5Eu melts congruently at 847 °C which is isostructural to $CaCu_5$ [10,13,14]; the Cu_2Eu is a $CeCu_2$-structure intermediate phase formed peritectically from liquid at 597 °C [10,15,16]; and CuEu of orthorhombic FeB-type crystal structure melts incongruently at 537 °C, [10,17,18]; while $CuEu_2$ of Ca_2Cu structure incongruently melts at 442 °C [10].

The heat of mixing of liquid in copper-rich concentration region was measured using isoperibolic calorimetry [9]. It was found that the minimum enthalpy of mixing corresponds to the

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Table 1
Intermetallic crystal structure in Cu–Eu and Cu–Yb system

Phase	Composition, at.% Yb	Pearson symbol	Space group	Prototype	Reference
Cu_5Eu	∼16.67	*hp*6	*P*6/*mmm*	$CaCu_5$	[13,14]
Cu_2Eu	∼33.3	*oI*2	*Imma*	$CeCu_2$	[15,16]
CuEu	∼50	*oP*8	*Pnma*	FeB	[17,18]
$CuEu_2$	∼66.67	*oP*12	*Pnma*	Ca_2Cu	[10]
Cu_5Yb	∼16.67	*hP*6	*P*6/*mmm*	$CaCu_5$	[21]
Cu_2Yb	∼33.3	*oI*12	*Imma*	$CeCu_2$	[22]
CuYb	∼50	*oP*8	*Pnma*	FeB	[19]

compound Ce_5Eu [9]. Other thermodynamic properties of the Cu–Eu system were not reported.

2.2. Cu–Yb system

The experimental information of the Cu–Yb system is limited. The phase diagram was studied by Iandelli et al. [19] based on differential thermal analysis (DTA), metallography and X-ray diffraction. On the basis of lattice parameter measurements, Iandelli et al. [19] determined that there is no appreciable terminal solid solubility in the Cu–Yb system. Later, Drits et al. [20] reported that the maximum solid solubility of Yb in Cu is ∼0.07 wt% (∼0.03 at.%) at the eutectic temperature. Of the five intermediate phases reported by Iandelli et al. [19], only Cu_9Yb_2 melts congruently at 937 °C whereas Cu_5Yb, Cu_7Yb_2, Cu_2Yb and CuYb form incongruently at 879, 825, 757 and 628 °C respectively [21,22]. The (βYb) → (γYb) transformation temperature is accepted from Dinsdale et al. [8] as 760 °C, which is 35 °C lower than that reported by Beaudry et al. [11]. The crystal structure of the three intermediate phases are listed in Table 1.

Heats of mixing of liquid in copper-rich concentration region were measured using isoperibolic calorimetry [9]. It was found that heats of mixing are negative [9].

3. Thermodynamic model

The lattice stabilities for element Cu, Eu and Yb are referred to Dinsdale [8]. An ordinary substitutional solution model is employed to describe liquid, BCC and FCC terminal solution. The molar Gibbs energy of a solution phase Φ (Φ = Liquid, FCC, BCC) can be represented as a sum of the weighted Gibbs energy for the pure components, the ideal entropy term describing a random mixing of the components, and the excess Gibbs energy describing the degree of deviation from ideal mixing, i.e.

$$G_m^{\Phi} = \sum x_i\, {}^0G_i^{\Phi} + RT \sum x_i \ln(x_i) + {}^EG_m^{\Phi} \tag{1}$$

where G_m^{Φ} is the molar Gibbs energy of a solution phase Φ, ${}^0G_i^{\Phi}$ the Gibbs energy of pure element i, x_i the mole fraction of component i (i = Cu, Eu or Cu, Yb), R the gas constant, T the temperature, ${}^EG_m^{\Phi}$ the excess Gibbs energy. For ordinary substitutional solution, the excess Gibbs energy of phase Φ can be written in the form of a Redlich–Kister polynomial as follows:

$${}^EG_m^{\Phi} = x_{\mathrm{Cu}} x_{\mathrm{M}} \sum_{j=0}^{n} {}^{(j)}L_{\mathrm{Cu,M}}^{\Phi} (x_{\mathrm{Cu}} - x_{\mathrm{M}})^j. \tag{2}$$

Here M is either Eu or Yb; ${}^{(j)}L_{\mathrm{Cu,M}}^{\Phi}$ an interaction parameter and can be expressed as temperature dependent as follows:

$${}^{(j)}L_{\mathrm{Cu,M}}^{\Phi} = A_j + B_j T + C_j T \ln(T) \tag{3}$$

where A_j, B_j and C_j are model parameters to be optimized.

The intermetallic phases have been treated as stoichiometric phases Cu_pM_q. Due to lack of heat capacity C_p and according to the Neumann–Kopp rule, the Gibbs energy of Cu_pM_q can be formulated as:

$$G_{\mathrm{Cu}_p\mathrm{M}_q} = \frac{p}{p+q}\, {}^0G_{\mathrm{Cu}}^{\mathrm{FCC}} + \frac{q}{p+q}\, {}^0G_{\mathrm{M}}^{S} + A + BT \tag{4}$$

where S is the stable phase of M at 298.15 K and S represents body-centered cubic (BCC) for Eu, face-centered cubic (FCC) for Yb. A and B are the adjustable parameters being optimized in the present work.

4. Results and discussion

On the basis of lattice stabilities cited from Dinsdale [8], the optimization of these two systems is carried out by using the Parrot modules in the Thermo-Calc program developed by Sundman et al. [23]. Thermodynamic parameters of all phases in these two systems are obtained and listed in Table 2. All experimental data including phase diagrams and thermodynamic properties are reasonably reproduced. It is reported that associates are often formed in the liquid states of the RE–TM alloys [25]. There is no direct evidence to confirm the obvious temperature dependence of enthalpy in those binaries, but in most of the other Cu–Re systems, this temperature dependence is significant, such as Cu–Dy [9,24], Cu–Ce [25], Cu–La [26,27], Cu–Gd [24,27,28] and so on. So it is believed that temperature dependence of the enthalpy of mixing is also obvious in the Cu–Eu and Cu–Yb systems. In order to describe this associate's tendency and to simplify the model, a $T \ln T$ parameter is used in these two liquid phases. More detailed analysis is given below.

4.1. Cu–Eu system

No solubility is assumed in the BCC (Eu) solution, which is realized by assigning a large positive interaction parameter, i.e.

$${}^0L_{\mathrm{Cu,Eu}}^{\mathrm{BCC}} = 5000\ \mathrm{J/mol}. \tag{5}$$

In this work, the lattice stabilities of Eu in FCC (${}^0G_{\mathrm{Eu}}^{\mathrm{FCC}} - \mathrm{GHSER_{EU}}$) is +180 000, which is assessed by Du et al. [29] and the interaction parameter of FCC solution phase is just given a large positive, i.e.

$${}^0L_{\mathrm{Cu,Eu}}^{\mathrm{FCC}} = 5000\ \mathrm{J/mol}. \tag{6}$$

Table 2
Thermodynamic parameters of the Cu–Eu and Cu–Yb system

System	Phase	Thermodynamic parameters
Cu–Eu	Liquid	${}^0L_{\mathrm{Cu,Eu}} = -25\,402.860 + 89.739T - 11.941T\ln(T)$ ${}^1L_{\mathrm{Cu,Eu}} = -15\,903.788 + 12.691T$ ${}^2L_{\mathrm{Cu,Eu}} = -11\,787.219$
	BCC	${}^0L^{\mathrm{BCC}}_{\mathrm{Cu,Eu}} = +30\,000$
	FCC	${}^0G^{\mathrm{FCC}}_{\mathrm{Eu}} - \mathrm{GHSER_{EU}} = +180\,000$ ${}^0L^{\mathrm{FCC}}_{\mathrm{Cu,Eu}} = +5000$
	Cu_5Eu	$G^{\mathrm{Cu_5Eu}} = 0.8333{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.1667{}^0G^{\mathrm{BCC}}_{\mathrm{Eu}} - 4363.535 - 1.237T$
	Cu_2Eu	$G^{\mathrm{Cu_2Eu}} = 0.6667{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.3333{}^0G^{\mathrm{BCC}}_{\mathrm{Eu}} - 5968.325 + 0.235T$
	CuEu	$G^{\mathrm{CuEu}} = 0.5{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.5{}^0G^{\mathrm{BCC}}_{\mathrm{Eu}} - 6135.327 + 1.137T$
	$CuEu_2$	$G^{\mathrm{CuEu_2}} = 0.3333{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.6667{}^0G^{\mathrm{BCC}}_{\mathrm{Eu}} - 5912.367 + 3.156T$
Cu–Yb	Liquid	${}^0L_{\mathrm{Cu,Yb}} = -41\,356.325 + 33.75T - 7.201T\ln(T)$ ${}^1L_{\mathrm{Cu,Yb}} = -30\,258.324 + 15.025T$
	FCC	${}^0L^{\mathrm{FCC}}_{\mathrm{Cu,Yb}} = +10\,000$ ${}^1L^{\mathrm{FCC}}_{\mathrm{Cu,Yb}} = -10\,000$
	Cu_5Yb	$G^{\mathrm{Cu_5Yb}} = 0.8333{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.1667{}^0G^{\mathrm{FCC}}_{\mathrm{Yb}} - 15\,339.485 + 2.2025T$
	Cu_9Yb_2	$G^{\mathrm{Cu_9Yb_2}} = 0.8182{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.1818{}^0G^{\mathrm{FCC}}_{\mathrm{Yb}} - 16\,248.917 + 2.0060T$
	Cu_7Yb_2	$G^{\mathrm{Cu_7Yb_2}} = 0.7778{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.2222{}^0G^{\mathrm{FCC}}_{\mathrm{Yb}} - 16\,841.985 + 1.4725T$
	Cu_2Yb	$G^{\mathrm{Cu_2Yb}} = 0.6667{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.3333{}^0G^{\mathrm{FCC}}_{\mathrm{Yb}} - 17\,792.410 + 5.7701T$
	CuYb	$G^{\mathrm{CuYb}} = 0.5{}^0G^{\mathrm{FCC}}_{\mathrm{Cu}} + 0.5{}^0G^{\mathrm{FCC}}_{\mathrm{Yb}} - 14\,796.220 - 1.4309T$

Note: Gibbs energies are expressed in J/mol. Lattice stabilities of elements Cu, Eu and Yb are referred to Dinsdale [8].

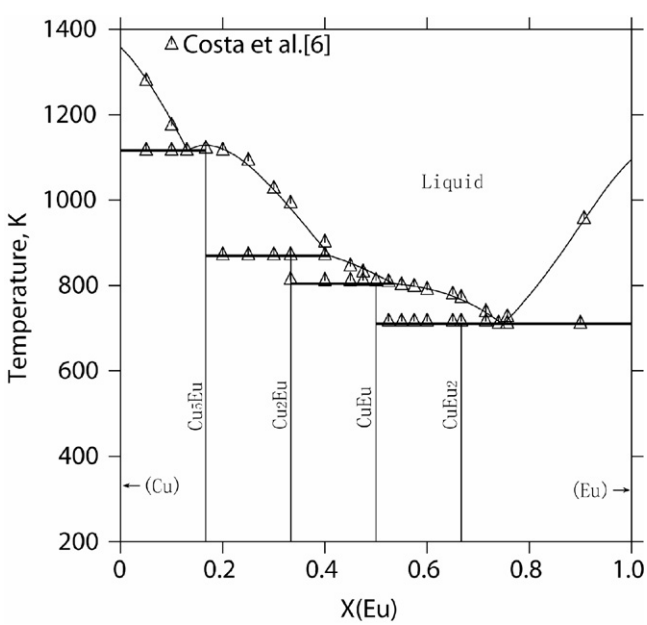

Fig. 1. The calculated Cu–Eu phase diagram with experimental data [10].

Table 3
Invariant reactions in the Cu–Eu system

Reaction	Composition of liquid (at.% Cu)	T (K)	Reaction type	Source
$L \Leftrightarrow FCC(Cu) + Cu_5Eu$	88.24 87	1109 1115	Eutectic Eutectic	This work [10]
$L \Leftrightarrow Cu_5Eu$	83.33	1122 1120	Congruent Congruent	This work [10]
$L + Cu_5Eu \Leftrightarrow Cu_2Eu$	59.19 59	869 870	Peritectic Peritectic	This work [10]
$L + Cu_2Eu \Leftrightarrow CuEu$	48.90 49	802 810	Peritectic Peritectic	This work [10]
$L + CuEu \Leftrightarrow CuEu_2$	26.49 26.7	720 715	Peritectic Peritectic	This work [10]
$L \Leftrightarrow BCC(Eu) + CuEu_2$	25.97 26	717 710	Eutectic Eutectic	This work [10]

The thermodynamic data of the liquid (the reference states are pure liquid Cu and Eu) and the liquidus of terminal solution are first used to optimize simultaneously the binary interaction parameters of the liquid. The parameters of various intermetallic phases are then optimized through fitting the phase diagram data, especially the invariant reactions. All the parameters were finally evaluated together to give a reasonable description of this system. Fig. 1 shows the calculated phase diagram of Cu–Eu system compared with experimental data. Combined with invariant reactions (as listed in Table 3), it is now clear that there is good agreement between the calculated values and the experimental phase diagram data. Within the experimental errors, the present calculation of the mixing enthalpy looks reasonably good for Cu-rich side, as shown in Fig. 2. Partial mixing enthalpies of Eu in the Cu–Eu liquid alloys at 1480 K are further calculated in comparison with the

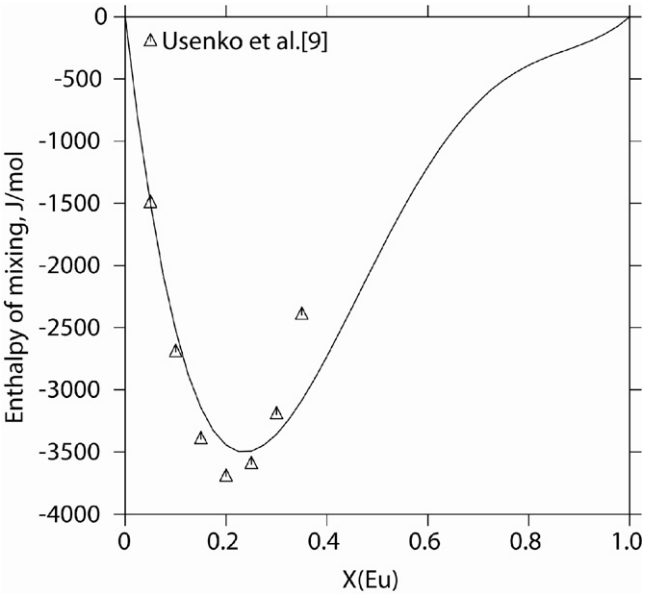

Fig. 2. Calculated enthalpy of mixing of liquid at 1480 K in Cu–Eu system with the experimental data [9] (the reference states are liquid Cu and liquid Eu).

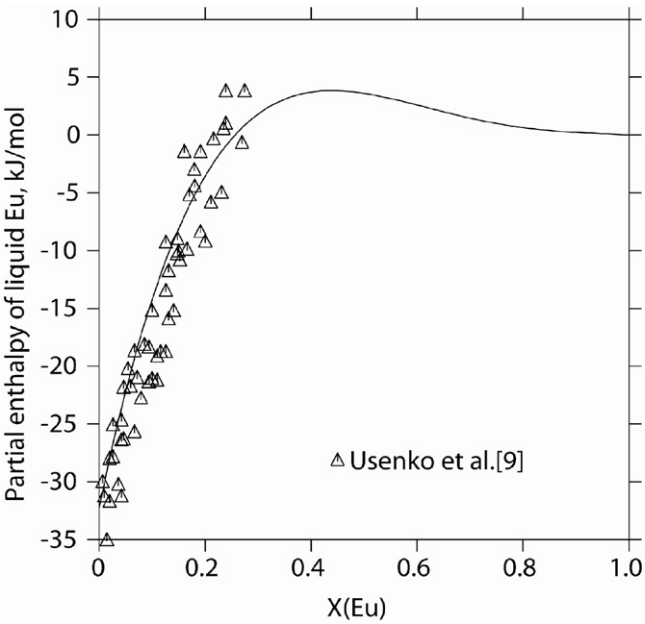

Fig. 3. Calculated partial enthalpy of Eu in liquid Cu–Eu alloys at 1480 K in comparison with experimental data [9].

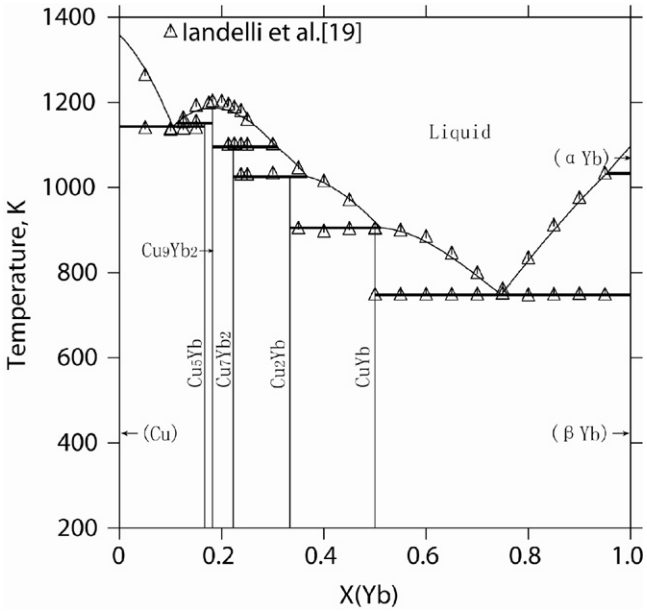

Fig. 4. The calculated Cu–Yb phase diagram with experimental data [19].

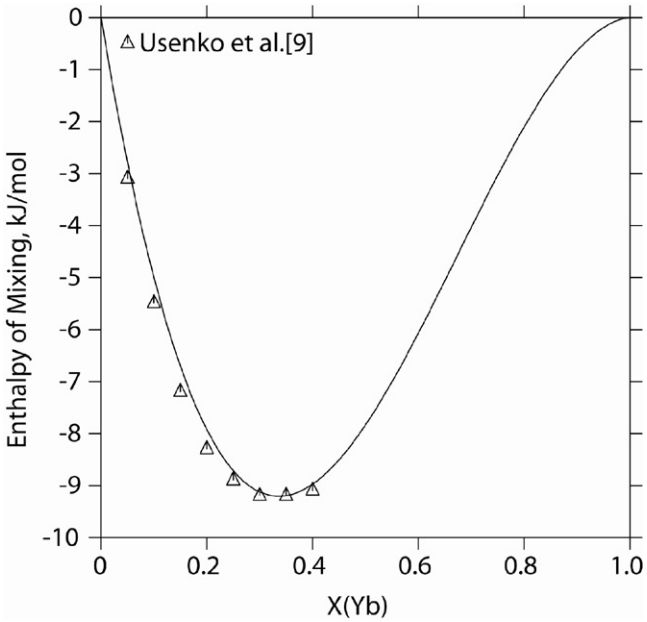

Fig. 5. Calculated enthalpy of mixing of liquid at 1453 K in Cu–Yb system with the experimental data [9] (the reference states are liquid Cu and liquid Yb).

experimental data, as shown in Fig. 3. It is demonstrated that the experimental data of thermodynamic properties can be well described by the present calculation with regard to experimental errors.

4.2. Cu–Yb system

Like in the Cu–Eu system, we conduct the assessment of the Cu–Yb system. Fig. 4 shows the calculated Cu–Yb equilibrium phase diagram with experimental values used in the optimization. Table 2 gives the optimized parameters for this system. The calculated phase diagram is in good agreement with that reported by Iandelli et al. [19]. All the assessed and experimental invariant equilibria in the Cu–Yb system are listed in Table 4. An agreement within 7 K between the temperature of calculated and experimentally determined invariant reactions has been obtained except for the congruent temperature of Cu_9Yb_2 phase, where the disagreement is 11 K. This reaction temperature, which is 1210 K, cannot be thermodynamically reproduced well despite great care taken in optimization, the phase equilibria measurements about this region need further clarification. Fig. 5 shows the calculated enthalpy of mixing

Table 4
Invariant reactions in the Cu–Yb system

Reaction	Composition of liquid (at.% Cu)	T (K)	Reaction type	Source
BCC(Yb) ⇔ FCC(Yb)	0	1033	Allotropic	This work
		1033	Allotropic	[8]
L ⇔ FCC(Yb) + CuYb	25.5	750	Eutectic	This work
	26.5	745	Eutectic	[19]
L + Cu_2Yb ⇔ CuYb	47.39	901	Peritectic	This work
	49	901	Peritectic	[19]
L + Cu_7Yb_2 ⇔ Cu_2Yb	62.08	1028	Peritectic	This work
	64	1030	Peritectic	[19]
L + Cu_9Yb_2 ⇔ Cu_7Yb_2	68.39	1098	Peritectic	This work
	70	1098	Peritectic	[19]
L ⇔ Cu_9Yb_2	81.82	1195	Congruent	This work
		1210	Congruent	[19]
L + Cu_9Yb_2 ⇔ Cu_5Yb	88.74	1151	Peritectic	This work
	88	1150	Peritectic	[19]
L ⇔ FCC(Cu) + Cu_5Yb	89.81	1139	Eutectic	This work
	90	1132	Eutectic	[19]

of liquid in Cu–Yb system at 1453 K with the experimental data [9]. From Figs. 4 and 5, we can see that the calculated phase diagram and thermodynamic properties of Cu–Yb system agree well with the experimental data.

5. Conclusions

The Cu–Eu and Cu–Yb binary systems have been assessed thermodynamically based on reported experimental data of phase diagrams and thermodynamic properties. Reasonable agreement between calculated and experimental data has been realized and thermodynamic parameters for various phases in these two binary systems are obtained.

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